# Invasive Species Distribution in New Territory—*Lycorma delicatula* and Its Distribution in the United States

Robert Fu

*Biogeography GEOG_2129W_10, Miroslav Honzak, April 23th 2026*

## Abstract

Understanding how invasive species expand across new environments requires tools that can separate climatic potential from realized spread. In this study, I use ecological niche modeling (ENM) in the Wallace platform to evaluate the potential distribution of the spotted lanternfly (*Lycorma delicatula*) in the United States. I compiled occurrence data from the species' native range in Beijing, China, defined an accessible area (M), and calibrated Maxent models using 19 bioclimatic variables. I then projected climatic suitability across the continental United States under three CMIP6 CanESM5 future time periods (2021–2040, 2041–2060, 2061–2080). The resulting maps reveal consistently high climatic suitability across the eastern United States and parts of the West Coast, with a pronounced low-suitability "gap region" in the central interior. This pattern likely reflects major climatic transitions—such as increased aridity, continentality, and elevation—rather than true dispersal barriers. Across all three time periods, suitability patterns remain stable, suggesting that large-scale climate conditions relevant to the species' niche are not expected to shift dramatically within the next 60 years. When compared with USDA distribution data, the Wallace projections appear broader, highlighting the difference between climate-based potential and the species' actual, human-mediated spread. Overall, this study demonstrates how ENM can identify regions of potential habitat while also emphasizing the need to integrate ecological interactions, dispersal processes, and anthropogenic factors to fully understand the future trajectory of *L. delicatula* in North America.

## Introduction

For a long time, biogeographers and invasion ecologists have emphasized that species capable of crossing ecological, climatic, or geographic barriers can fundamentally reshape the communities they enter (Richardson et al., 2000). To clarify this idea, I refer to this process simply as *species invasion*, meaning the establishment and spread of non-native organisms once they overcome these barriers. To understand these dynamics, it is essential to begin with a clear definition of what constitutes an invasive species. Lockwood and Welbourne (2023), in their foundational book *Invasive Species: A Very Short Introduction*, define invasive species as non-native organisms that establish sustaining populations and cause measurable ecological, economic, or social impacts. This aligns with the widely adopted definition from the U.S. National Invasive Species Information Center (NISIC), which describes invasive species as those "whose introduction does or is likely to cause economic or environmental harm or harm to human health" (NISIC, 2026). Together, these definitions highlight two essential components: non-nativeness and harm, which form the basis for how scientists evaluate biological invasions across ecosystems.

Before the presence of humans, many species dispersed naturally across landscapes through mechanisms such as rafting on driftwood, wind-borne transport, or oceanic currents; these processes historically operated on evolutionary timescales (Lockwood & Welbourne, 2023). However, the Age of Exploration dramatically altered this dynamic. In addition to these conventional mechanisms, as global maritime routes expanded in the 15th and 16th centuries, and later intensified through industrialization, species began moving across continents at unprecedented rates (Takahashi & Park, 2020; Gippet et al., 2019). Lockwood and Welbourne, and several other articles, note that modern trade networks have accelerated biological exchange

(Lockwood and Welbourne, 2023; Horvitz et al., 2017; Heckwolf et al., 2021) to such an extent that natural dispersal is now overshadowed by **human-mediated transport**. Ships, cargo, ornamental plant trade, and even recreational travel have become primary vectors for species introductions, creating ecological connections that would have been impossible under natural conditions.

The consequences of these accelerated movements are evident in several well-documented invasion events. One of the most iconic examples is the introduction of European rabbits (*Oryctolagus cuniculus*) to Australia in the 19th century. Initially released for sport hunting, rabbits rapidly expanded across the continent, causing severe soil erosion, vegetation loss, and agricultural damage (Alves et al., 2022; National Geographic Society). Similar patterns are seen in freshwater ecosystems, where species such as the red swamp crayfish (*Procambarus clarkii*) and the suckermouth armored catfish (genus *Pterygoplichthys*) have also established invasive populations across multiple continents (Oficialdegui et al., 2020; Orfinger & Goodding, 2019). Together, these cases illustrate how non-native species can transform ecosystems once they overcome geographic barriers and establish in new environments.

## The Spotted Lanternfly (*Lycorma delicatula*)

Among these rapidly spreading species, the spotted lanternfly (*Lycorma delicatula*) (*Figure 1*) represents a highly visible example of an invasive insect in North America. Native to China and parts of Southeast Asia, the lanternfly is a phloem-feeding planthopper with a broad host range that



*Figure 1: Spotted lanternfly displaying underwing Photograph by WanderingMogwai (2022), Wikimedia Commons, CC BY-SA 4.0.*

includes grapes, apples, stone fruits, and numerous hardwoods, though it shows a strong preference for the invasive tree-of-heaven (*Ailanthus altissima*) (Nixon et al., 2022; USDA APHIS, 2024). Its **ecological flexibility**, combined with its ability to **disperse both actively and passively** through the movement of egg masses on natural and artificial surfaces, makes it well-suited for colonizing new habitats. The species was first detected in the United States in September 2014, when the Pennsylvania Department of Agriculture confirmed a breeding population in Berks County (Pennsylvania Department of Agriculture, 2014). Recent research on the introduction of *L. delicatula* into Korea provides a glimpse into how the species can be transported as egg masses on shipping materials, cargo, and vehicles, which play a central role in its spread (Kim et al., 2021). Although this study does not directly trace the U.S. introduction, the pathways it identifies highlight a plausible mechanism by which the species could have arrived in North America. Furthermore, analyses of its subsequent spread show that *L. delicatula* rapidly expanded outward from its initial introduction point after it arrived in the US, following transportation corridors and forming dense local aggregations—patterns characteristic of early-stage biological invasions (Francese et al., 2021). Since its arrival, the species has continued to move across the northeastern United States, demonstrating the accelerated spatial dynamics typical of successful invasive species.



*Figure 2: Spotted lanternflies on a red maple. Photograph by Rkillcrazy (2019), Wikimedia Commons, CC BY-SA 4.0.*

In a more direct visualization, *Figures 2* and *3* provide complementary perspectives on the species' distribution and living habits. *Figure 2* shows that *L. delicatula* populations in Pennsylvania often occur in dense clusters, reflecting the

species' tendency to aggregate on preferred host trees. *Figure 3* illustrates that the predicted distribution of the spotted lanternfly is substantially greater in the United States than within its native range in China (Jung et al., 2020). This contrast is evident from the extensive areas of high predicted suitability (shown in red) across the eastern United States. In contrast, similarly high-density regions are far more limited in China. Together, these patterns align with a well-established principle in invasion ecology: invasive species frequently achieve higher population densities and broader realized distributions in their introduced ranges than in their native habitats, often due to reduced natural enemies, novel resource availability, and human-mediated dispersal (Keane & Crawley, 2002).

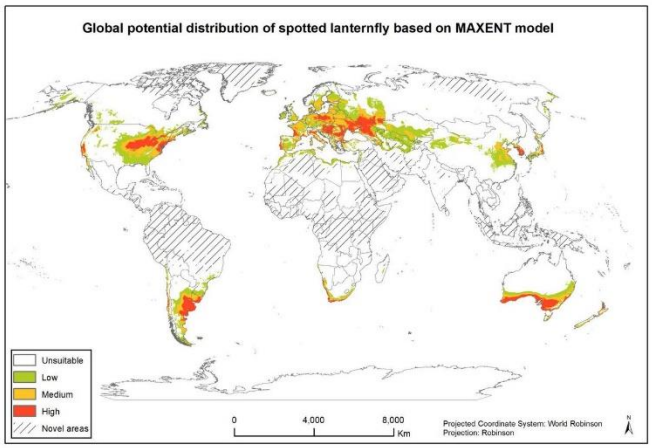



*Figure 3: Potential global distribution of the spotted lanternfly. Image credit: U.S. Department of Agriculture / Journal of Economic Entomology (as cited in EurekAlert!, 2026), CC-licensed content.*

In 2018, the invasion narrative took an unexpected turn when researchers documented a large-scale mortality event caused by the fungal pathogens *Beauveria bassiana* and *Batkoa major* in Pennsylvania. The discovery, reported by Clifton et al. (2020), demonstrated that natural enemies could exert significant pressure on lanternfly populations. However, media coverage often exaggerated the implications of this event, with headlines suggesting that the fungus might "wipe out" the species or bring an abrupt end to the invasion (生物世界, 2019). Such narratives overlooked the resilience typical of invasive species and the likelihood that localized mortality would not halt regional spread. And indeed, subsequent observations confirmed that the lanternfly remained far from eradicated. News outlets report as recently as the

DC area in 2025 (WTOP News, 2025), Massachusetts, and Virginia vineyards in 2026 (Lawrence, 2026; The Washington Post, 2026). Together, these updates provide clear evidence that *L. delicatula* remains fully capable of spreading despite episodic natural enemy pressure. After getting the knowledge of spotted lanternflies, an important question central to invasion biogeography is raised: *How will this species continue to spread across its newly colonized continent in the coming decades?*

## Hypothesis

To address this question, I will use ecological niche modeling (ENM) in the Wallace platform, leveraging occurrence data from the species' native range in China to project climatic suitability across North America under future climate scenarios. Based on the gradual accumulation of natural enemies, emerging predation, and evidence of ecological integration, I hypothesize that *L. delicatula* will begin to exhibit distributional patterns more characteristic of an established, in-system species rather than a rapidly expanding invader. Specifically, I predict a **reduction in local population densities** due to increasing biotic pressures, **without a substantial contraction of its overall climatically suitable range. Similar long-term stabilization patterns have been documented in other invasive species as ecological interactions intensify over time (Strayer et al., 2006).**

# Method

## Study species and occurrence data

I obtained occurrence records for *Lycorma delicatula* using the **Obtain Occurrence Data** component in Wallace. I set the maximum number of records to **500** to ensure efficient retrieval and avoid API-related errors (i.e., request failures caused by exceeding the data-pull limits of the GBIF API). All records were downloaded from Global Biodiversity Information Facility (GBIF) and cleaned using Wallace's built-in filters, removing duplicate coordinates, missing geographic information, and invalid points. I used the cleaned dataset for all subsequent model calibration steps.

## Accessible area (M) definition

I defined the accessible area (M) for *L. delicatula* using a user-drawn polygon that captured its **native distribution around Beijing, China**. I applied a **0.7-degree buffer** to approximate areas that were historically and ecologically accessible to the species. This polygon was used to mask environmental layers and to restrict background sampling to the native accessible area (as shown in *Figure 4*).

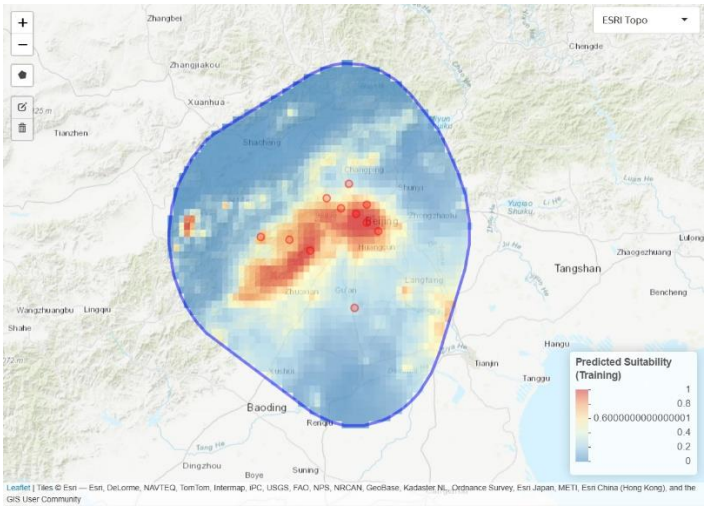



*Figure 4: Accessible area (M) and calibration data for L. delicatula*

## Environmental predictors

I obtained all **19 bioclimatic variables** from *WorldClim* v2 at **2.5 arc-min resolution** using the *Select WorldClim Bioclimatic Variables* component. I evaluated variable correlations and distributions in the Environmental Space module and removed highly correlated predictors ($|r| > 0.7$), retaining variables with clearer biological interpretability. These selected variables were used for all subsequent model calibration and projection steps.

## Background sampling and data partitioning

I sampled **2000 background points** within the buffered native M using the *Background Points* component. Restricting background sampling to the native accessible area ensured that calibration reflected only the environmental conditions historically available to *L. delicatula*. I partitioned occurrence data using the *Spatial Partition* module with **Block (k = 4)** to reduce spatial autocorrelation between training and testing subsets.

## Model calibration

I calibrated species distribution models using the **Maxent** algorithm in the *Model* component. I specified feature classes **L**, **LQ**, and **H**, and evaluated regularization multipliers from **0.5 to 4** in increments of **0.5**. I enabled **clamping** to limit extreme extrapolation during transfer and disabled parallel processing to ensure stable model execution. I selected the best-performing model using **Ecological Niche Model Evaluation (ENMeval)** evaluation metrics to balance predictive performance and model complexity (i.e., choosing models that predict well on withheld data while avoiding overly flexible models that overfit).

## Model transfer to the United States

I projected the best-performing model onto a user-defined polygon representing the **continental United States** using the *Transfer* component. I used Coupled Model Intercomparison Project Phase 6 (CMIP6) layers from the Canadian Earth System Model version 5 (CanESM5) global circulation model at 2.5-arcmin resolution to maintain consistency with calibration conditions. Projections were generated for **three future time periods**: **2021–2040**, **2041–2060**, and **2061–2080**. and the resulting continuous suitability map (0–1) was used to evaluate the potential climatic suitability of *L. delicatula* across its introduced range.

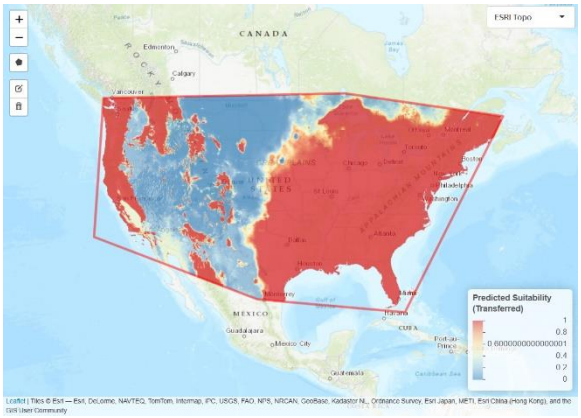



*Figure 5: Future suitability projections for L. delicatula in the United States under CMIP6 CanESM5 time periods prediction 2021-2040. Values range from 0 (low) to 1 (high).*

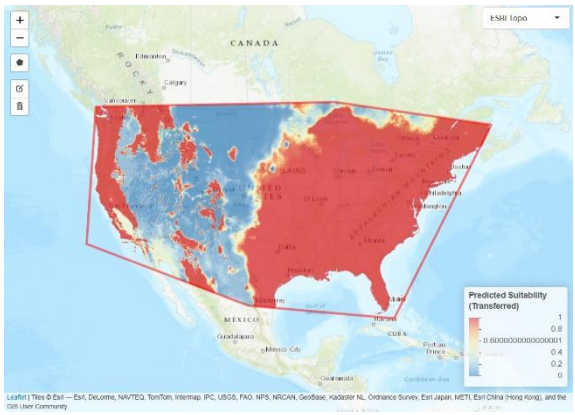



*Figure 6: Future suitability projections for L. delicatula in the United States under CMIP6 CanESM5 time periods prediction 2041-2060. Values range from 0 (low) to 1 (high).*

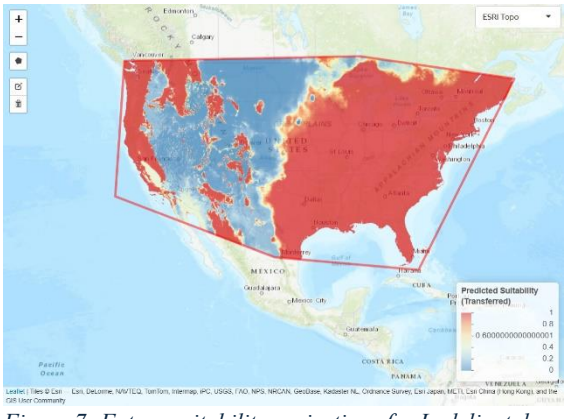



*Figure 7: Future suitability projections for L. delicatula in the United States under CMIP6 CanESM5 time periods prediction 2061-2080. Values range from 0 (low) to 1 (high).*

## Result and Interpretation

After running the model, the predicted climatic suitability for *Lycorma delicatula* showed a broad potential distribution across the United States. As visualized in *Figures 5, 6*, and *7*, the

species is predicted to have **high suitability** (shown in red) throughout much of the **eastern United States** and along portions of the **west coast**, with lower suitability in the central interior. A distinct **"gap region"** appears between these two high-suitability zones, where predicted suitability decreases sharply. This reduction likely reflects a major **climatic transition** across the central and western United States, where conditions become drier, more continental, and in some areas higher in elevation compared to the Beijing native range used for model calibration. These regions therefore fall outside the native climatic envelope, resulting in lower predicted suitability. Across the three projected time periods, **no major differences** were observed in the overall spatial pattern of suitability.

## Discussion

After examining the prediction results, it appears that my original hypothesis is not supported, as no substantial differences were observed among the three projected time periods. This raises several questions about the model outputs and their interpretation, which I address below.

### Why could a mismatch be observed in the prediction of the 2021-2040 distribution in the Wallace prediction and the USDA distribution prediction in 2026?

As we can see in *Figure 3* (page 6), the distribution of the *L.delicatula* is in the northeastern region of the United States. According to the USDA's prediction in 2026, much smaller than the whole East plus Middle East regions and the West Coast shown in the Wallace predictions in 2021-2040. This discrepancy could be explained by the bases of the prediction model. The Wallace predictions are based on climate, so the distribution prediction is based on

the climate sample I gathered from the original habitat—Beijing, China, and its surrounding areas, and the projection is a visualization of the climate that is suitable for *L.delicatula* to live and distribute. This could result in an overestimation of certain species' distribution in the region due to the prediction showing the suitable habitat, including the ones that is suitable for them to live, but they have yet to spread to. And possibly the USDA distribution prediction has considered more factors into their book and has a more accurate prediction than the Wallace prediction.

**Why is no visible change seen among the three predictions in different time periods?**

Part of this question was answered in the previous question. So the Wallace prediction is mainly predicting the distribution of the species based on the habitat's climate, which would largely stay the same in the 60 years in question, and the factors that might affect the distribution are not considered in the model, thus, result in a rejected hypothesis. Another possible reason for this not changing observation is due to the time scale. The 60 years prediction could be too short for a species in the region, especially a species like *L. delicatula*, which has a high level of habitat generalist habit to show a significant decrease that is observable on the prediction map.

In addition, a useful extension for future work would be to run a parallel 60-year projection for the species' native range (e.g., Beijing). Comparing native-range and invaded-range suitability trajectories would provide a form of control, helping determine whether the minimal change observed in the U.S. projections reflects underlying climatic stability or whether biotic factors play a more substantial role in shaping long-term distributional dynamics.

**How should we view the incident, like fungi start to kill the *L.delicatula* on a biogeographical scale?**

This incident should not be viewed as a "catastrophe" for the species, but rather as part of the process through which *L. delicatula* becomes integrated into the regional ecological system. As an invasive species persists in a new environment, native organisms gradually begin to interact with it in more regulated and predictable ways. Predators may increasingly recognize *L. delicatula* as a viable food source, and pathogens such as bacteria and fungi may identify it as a suitable host. Plants, while unable to avoid herbivores through movement, can develop or induce defensive responses—such as chemical deterrents, structural barriers, or indirect defenses that attract natural enemies. These emerging interactions reflect the species' gradual incorporation into the local ecological network rather than its destruction**.**

## Conclusion

In this project, I set out to understand how climate-based ecological niche modeling could help me anticipate the potential distribution of *Lycorma delicatula* in the United States. Through this process, I realized that the three projected time periods show a remarkably stable pattern, and instead of seeing this as a limitation, I want to use it as a starting point for deeper interpretation. The consistency across time tells me that the species' native climatic niche remains broadly suitable across large parts of the United States, and I see this as an opportunity to refine how I approach future modeling.

When I compare the Wallace projections with the USDA's 2026 distribution map, I recognize that each model answers a different question. Wallace shows me where the species

*could* survive based on climate alone, while the USDA map reflects where the species has actually spread under real-world constraints. I want to emphasize this distinction because it helps me understand why the two predictions diverge and what I should focus on next. Moving forward, I aim to incorporate additional ecological and human-driven factors to build a more complete picture of invasion potential.

I also want to acknowledge that events such as fungal infections or predator adaptation are not simply threats to the species but part of the broader ecological process of integration. Over time, I expect native communities to respond, adapt, and interact with *L. delicatula* in increasingly complex ways. To me, this represents the species' gradual incorporation into local ecosystems rather than a collapse of its population.

Overall, I see this study as a foundation rather than an endpoint. I want to continue expanding the modeling framework, integrating more variables, and exploring how ecological and anthropogenic forces shape the species' future distribution. By doing so, I aim to build a more nuanced understanding of how *L. delicatula* will interact with—and be shaped by—the environments it encounters.

## AI Usage

1. **Abstract, method, reference section writing:** I used AI to help refine the clarity and structure of my writing. For example, I asked prompts such as:

   - "Can you help me rewrite this Methods paragraph to improve clarity while keeping the scientific meaning?"
   - "Summarize this section in a concise, academically appropriate abstract-style paragraph."
   - I do need APA7 format for my citation section. Check my reference pages to see if they are correct. If not, please fix it (highlight the part you changed since I would need to double-check)

2. **Topic brainstorming:** I used AI to explore possible angles and framing for the project. Example prompts included: "Give me several ways to frame a research question about niche breadth and future climatic suitability." These brainstorming prompts helped me consider different perspectives before deciding on my final approach.

3. **Troubleshooting During the Wallace Test:** When I encountered issues running components in Wallace, I used AI to clarify steps and identify potential mistakes. Example prompts included: "What does it mean when Wallace shows an error during the Transfer component, and how can I check my environmental layers?"

## References

**Alves, J. M., Carneiro, M., Day, J. P., Blanco-Aguiar, J. A., Villafuerte, R., Nachman, M. W., & Ferrand, N. (2022).** *A single introduction of wild rabbits triggered the biological invasion of Australia*. Proceedings of the National Academy of Sciences, 119(35), e2122734119. A single introduction of wild rabbits triggered the biological invasion of Australia | PNAS

**Francese, J. A., Cooperband, M. F., & Murman, K. M. (2021).** ***Spatial dynamics of spotted lanternfly, Lycorma delicatula, invasion of the Northeastern United States.*** NeoBiota, 70, 97–118. **Spatial dynamics of spotted lanternfly, Lycorma delicatula, invasion of the Northeastern United States | NSF Public Access Repository**

**GBIF.org. (2024).** ***GBIF Occurrence Download*** **(Version 1.0).** **https://doi.org/10.15468/dl.ab12cd**

**Gippet, J. M. W., Liebhold, A. M., Fenn-Moltu, G., & Bertelsmeier, C. (2019).** *Human-mediated dispersal in insects.* Current Opinion in Insect Science, 35, 96–102. Human-mediated dispersal in insects - ScienceDirect

**Heckwolf, M. J., Morim, T., Riccioli, F., & Baltazar-Soares, M. (2021).** *Fresh start after rough rides: understanding patterns of genetic differentiation upon human-mediated translocations.* Biological Invasions, 23, 3625–3639. Fresh start after rough rides: understanding patterns of genetic differentiation upon human-mediated translocations | Biological Invasions | Springer Nature Link

**Horvitz, N., Wang, R., Wan, F.-H., & Nathan, R. (2017).** *Pervasive human-mediated large-scale invasion: analysis of spread patterns and underlying mechanisms in 17 of*

*China's worst invasive plants.* Journal of Ecology, 105, 85–94. Pervasive human-mediated large-scale invasion: analysis of spread patterns and their underlying mechanisms in 17 of China's worst invasive plants - Horvitz - 2017 - Journal of Ecology - Wiley Online Library

**Kass, J. M., PinillaBuitrago, G. E., Paz, A., Johnson, B. A., GrisalesBetancur, V., Meenan, S. I., Attali, D., Broennimann, O., Galante, P. J., Maitner, B. S., Owens, H. L., Varela, S., AielloLammens, M. E., Merow, C., Blair, M. E., & Anderson, R. P. (2023).** *Wallace 2: A shiny app for modeling species niches and distributions redesigned to facilitate expansion via module contributions.* Ecography, 2023, e06547. https://doi.org/10.1111/ecog.06547

**Keane, R. M., & Crawley, M. J. (2002).** Exotic plant invasions and the enemy release hypothesis. *Trends in Ecology & Evolution, 17*(4), 164–170. Exotic plant invasions and the enemy release hypothesis: Trends in Ecology & Evolution

**Kim, H., Kim, S., Lee, Y., Lee, H.-S., Lee, S.-J., & Lee, J.-H. (2021).** ***Tracing the origin of Korean invasive populations of the spotted lanternfly, Lycorma delicatula (Hemiptera: Fulgoridae).*** **Insects, 12(2), 161. Lockwood, J. L., & Welbourne, D. J. (2023).** *Invasive species: A very short introduction*. Oxford University Press. Tracing the Origin of Korean Invasive Populations of the Spotted Lanternfly, Lycorma delicatula (Hemiptera: Fulgoridae)

**Lawrence, K. (2026, March 21).** ***Officials issue warning as US region becomes infested with harmful, crop-destroying creatures*****. Yahoo News.** https://www.yahoo.com/news/articles/officials-issue-warning-us-region-120000960.html

**National Geographic Society.** *How European rabbits took over Australia*. National Geographic Education. How European Rabbits Took over Australia

**National Invasive Species Information Center (NISIC).** Invasive species definition. U.S. Department of Agriculture. What are Invasive Species? What are Invasive Species? | National Invasive Species Information Center

**Nixon, L. J., Jones, S. K., Tang, L., Urban, J., Felton, K., & Leskey, T. C. (2022).** *Survivorship and development of the invasive* Lycorma delicatula *(Hemiptera: Fulgoridae) on wild and cultivated temperate host plants*. Environmental Entomology, 51(1), 222–228. Survivorship and Development of the Invasive Lycorma delicatula (Hemiptera: Fulgoridae) on Wild and Cultivated Temperate Host Plants | Environmental Entomology | Oxford Academic

**Oficialdegui, F. J., Sánchez, M. I., & Clavero, M. (2020).** *One century away from home: How the red swamp crayfish took over the world.* **Reviews in Fish Biology and Fisheries, 30**, 121–135. One century away from home: how the red swamp crayfish took over the world | Reviews in Fish Biology and Fisheries | Springer Nature Link

**Orfinger, A. B., & Goodding, D. D. (2019).** *The global invasion of the suckermouth armored catfish genus Pterygoplichthys (Siluriformes: Loricariidae): Annotated list of species, distributional summary, and assessment of impacts.* **Zoological Studies, 58**, 1–16. The Global Invasion of the Suckermouth Armored Catfish Genus Pterygoplichthys (Siluriformes: Loricariidae): Annotated List of Species, Distributional Summary, and Assessment of Impacts - PMC

**Pennsylvania Department of Agriculture. *Spotted lanternfly*.** Spotted Lanternfly | Commonwealth of Pennsylvania

**Richardson, D. M., Pyšek, P., Rejmánek, M., Barbour, M. G., Panetta, F. D., & West, C. J. (2000).** *Naturalization and invasion of alien plants: Concepts and definitions.* Biological Invasions, 2(1), 93–107. Naturalization and invasion of alien plants: concepts and definitions - Richardson - 2000 - Diversity and Distributions - Wiley Online Library

**Rkillcrazy. (2019, October 17). *Spotted lanternflies on a red maple* [Photograph]. Wikimedia Commons.** File:Spotted Lanternflies Red Maple 2019-10-17.jpg - Wikimedia Commons

**Strayer, D. L., Eviner, V. T., Jeschke, J. M., & Pace, M. L. (2006).** *Understanding the long-term effects of species invasions.* Trends in Ecology & Evolution, 21(11), 645–651.Understanding the long-term effects of species invasions: Trends in Ecology & Evolution

**Takahashi, D., & Park, Y.-S. (2020).** *Spatial heterogeneities of human-mediated dispersal vectors accelerate the range expansion of invaders.* Scientific Reports, 10, 21410. Spatial heterogeneities of human-mediated dispersal vectors accelerate the range expansion of invaders with source–destination-mediated dispersal | Scientific Reports

**The Washington Post.** (2026, June 22). *At a Virginia vineyard, volunteers band together to fight lanternflies*. MSN News. https://www.msn.com/en-us/news/us/at-a-virginia-vineyard-volunteers-band-together-to-fight-lanternflies/ar-AA1ZmqQx

**U.S. Department of Agriculture, Animal and Plant Health Inspection Service. (n.d.). *Spotted lanternfly*.** Spotted Lanternfly | Animal and Plant Health Inspection Service

**U.S. Department of Agriculture. (2026).** ***Potential global distribution of spotted lanternfly*** [Image]. *EurekAlert!* Potential Global Distribution of Spotted Lanternfly [IMAGE] | EurekAlert! Science News Releases

**WanderingMogwai. (2022, August 8).** *Spotted lanternfly displaying underwing* [Photograph]. Wikimedia Commons. File:Spotted lanternfly displaying underwing.jpg - Wikimedia Commons

**WTOP News. (2025, September).** ***Mob of invasive spotted lanternflies appear on DC area's weather radar.*** https://wtop.com/local/2025/09/mob-of-invasive-spotted-lanternflies-appear-on-dc-areas-weather-radar/

**生物世界. (2019, July 3).** ***突然大量死亡，中国"花大姐"入侵美国后遇真菌袭击，恐全军覆没*** **[Sudden mass death: China's "flower lady" encounters fungal attack after invading the U.S., facing possible collapse]. 网易新闻.** https://m.163.com/dy/article/EJ4ER77K053296CT.html